

4 Cell membranes and transport

AS · 2026 examination · Concept guide and original exam-style practice

Understand the concept

A phospholipid bilayer has hydrophilic heads facing water and hydrophobic tails inward. Proteins support facilitated diffusion, active transport and signalling. Diffusion proceeds down a concentration gradient; osmosis is net water movement from higher to lower water potential through a partially permeable membrane. Active transport uses ATP to move substances against gradients.

Worked example: follow the reasoning

A root hair cell can absorb mineral ions even when their concentration is lower outside: a carrier uses ATP to transport ions inward against the gradient.

Quick check · multiple choice

A What directly drives active transport against a gradient?

[1]

- A Osmosis
- B Simple diffusion
- C Gravity
- D ATP use

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 Explain why a plant cell loses turgor in concentrated sucrose but may stay viable.

[3]

Answers and explanation

Quick check: D. ATP use

Question 1 [3 marks]

Outside solution has lower water potential [1]; water exits by osmosis through the partially permeable membrane [1]; membrane pulls from wall (plasmolysis), but wall remains intact and cell can recover [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Biology 9700, 2025–2027. <https://www.cambridgeinternational.org/Images/664560-2025-2027-syllabus.pdf>

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